

HAT-P-20b

R-1167

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_250207 · created 2026-07-31T03:02:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460713.67 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.056
Transit depth (Rp/Rs)^2	3.23 +/- 2.01 [%]
Orbital Inclination (inc)	84.88 +/- 2.38
Airmass coefficient 1 (a1)	1.05 +/- 1.09
Airmass coefficient 2 (a2)	0.14 +/- 1.47
Scatter in the residuals of the lightcurve fit is	127.41 %
Best Comparison Star	#2 - [250, 208]
Optimal Aperture	1.34
Optimal Annulus	4.25
Transit Duration (day)	0.044 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.18 $\pm$ 0.056 vs 0.1284 (deviation 0.67 σ)
Duration fit vs published	0.044 d vs 0.0775 d (deviation 0.64 σ)
Mid-time O-C	-7.2 min
Depth SNR	2.3
Residual scatter	127.41 %

Light curve

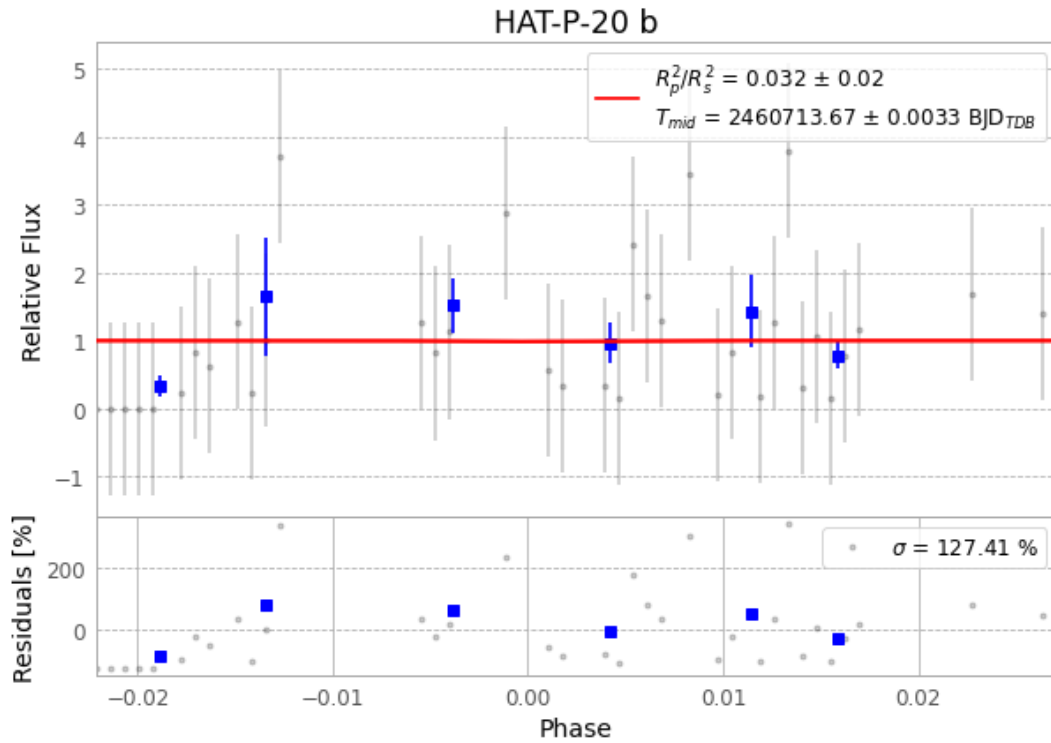


Figure 1 — FinalLightCurve_HAT-P-20 b_2025-02-06.png (SHA-256 463e2032cf98eb1c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [116, 151], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b355adbb13dec802...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), HATP-20250207022420.FITS → HATP-20250207055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-250207.log (d1b14357a1a8...), FinalLightCurve_HAT-P-20 b_2025-02-06.png (463e2032cf98...), FinalLightCurve_HAT-P-20 b_2025-02-06.csv (a21344fa8edb...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:02Z.